WEEK 5 · LECTURE L2 · UNIT II

Learning Management Systems

From course catalogue to skills platform, and what makes learning change behavior.

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi
DSEH001 · Digital Transformation in Human Resource Systems · M.Com. PGCF 2025 · Exam preparation edition

AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Apply andragogy, experiential learning and self-determination theory to digital learning design.
- Explain the forgetting curve and desirable difficulties, and their design implications.
- Distinguish LMS, LXP and skills platforms, using the Infosys Lex case.
- Critically evaluate Kirkpatrick's four levels, transfer of training, 70-20-10 and training ROI.

COURSE OUTCOMES

CLO 2 (learning systems) and CLO 4 (metrics and evidence).

THEORY 8

How adults learn: two theories set the design brief

Knowles: andragogy

Adults need to know why; they direct their own learning; experience is their main resource; they learn around real problems.

Kolb: experiential cycle (1984)

Concrete experience, reflective observation, abstract conceptualization, active experimentation.

EXAM LENS

Matching teaching to individual "learning styles" has no good evidence (Pashler et al., 2008). Citing this earns credit for criticality.

Why points and badges can backfire

Three basic needs

Deci and Ryan (1985); Ryan and Deci (2000): autonomy, competence and relatedness sustain intrinsic motivation.

Overjustification

Deci, Koestner and Ryan (1999), meta-analysis: expected tangible rewards undermined intrinsic motivation for interesting tasks; informational feedback did not.

Design consequences

Choice of path (autonomy), visible skill growth (competence), learning with peers (relatedness).

The forgetting curve and desirable difficulties

Principle	Finding	Design implication
Forgetting	Memory fades fast without review (Ebbinghaus, 1885)	Plan reviews, not single sessions
Spacing	Spread practice beats massed practice (Cepeda et al., 2006)	Short refreshers on days 2, 7 and 21
Retrieval practice	Testing beats rereading (Roediger and Karpicke, 2006)	Quiz before and after content
Interleaving	Mixed problems beat blocked ones (Rohrer and Taylor, 2007)	Mix related tasks in practice
Segmenting	Short learner-paced parts help novices (Mayer, 2009)	One task per short clip

EXAM LENS

Bjork (1994) called these "desirable difficulties": they slow apparent progress but improve long-term retention.

PLATFORMS

LMS, LXP and skills platform: three different logics

	LMS	LXP	Skills platform
Organizing unit	Course	Content experience	Skill
Who chooses	HR assigns	Learner browses	Role gap drives
Main metric	Completion	Engagement	Proficiency gain
Best use	Compliance	Self-directed professionals	Frontline reskilling

CASE · INFOSYS (2018)

Infosys Lex: digital learning at scale

What it is

Infosys's in-house learning platform (called Lex internally), launched in April 2018: mobile-first, with machine-learning recommendations and peer-created content.

Scale

Used by more than 114,000 Infosys employees within months of launch.

Then sold outside

In September 2018 Infosys began offering the platform to client firms as Infosys Wingspan.

EXAM LENS

Ask which organizing unit Lex chose and what that made measurable. Then ask what would not transfer to a frontline, multilingual, smartphone-only workforce.

Four levels of training evaluation

Level	Question	Typical evidence
4. Results	Did organizational outcomes change?	Attrition, service quality, productivity
3. Behavior	Do people do things differently at work?	Observed use, self-service transactions
2. Learning	Did they learn?	Assessment scores
1. Reaction	Did they value it?	Feedback forms

CRITIQUE

The four levels are not a causal chain

Alliger et al. (1997)

Meta-analysis: reactions correlated only weakly with learning and with on-the-job behavior. Judgments of usefulness did better than enjoyment.

Holton (1996)

The model is a taxonomy, not a theory: it names outcomes without explaining why one leads to another; context and motivation are left out.

A high Level 2 score licenses no claim about Level 3. Behavior must be measured directly.

THEORY 10B · BALDWIN AND FORD (1988); BLUME ET AL. (2010)

Transfer of training: the missing causal model

Inputs

Trainee ability and motivation; training design (practice, relevance); work environment (support, opportunity to use).

Learning

What is learned and retained.

Transfer

Used on the job and maintained. Blume et al.: work environment support and motivation relate to transfer.

EXAM LENS

A module cannot supply the manager who lets you use what you learned. Transfer is an organizational outcome, not a course outcome.

THEORY IN DEPTH · LOMBARDO AND EICHINGER (1996)

70-20-10: popular, but weakly evidenced

THE CORE IDEA

The claim: about 70% of learning comes from work, 20% from others and 10% from formal courses.

HOW IT WORKS

It pushes organizations to design learning into the job, with coaching and stretch tasks.

LIMITS AND CRITIQUE

The ratio comes from executive self-reports, not measurement; treat it as a reminder, not a fact.

IN PRACTICE

For a frontline associate, the shift and the buddy are most of the curriculum; the LMS supports them.

EXAM LENS

Use 70-20-10 only with its evidence caveat. Presenting it as a research finding loses marks.

THEORY IN DEPTH · PHILLIPS (1997)

Add a fifth level: return on investment

THE CORE IDEA

Phillips extended Kirkpatrick with a fifth level that compares monetary benefits with costs.

HOW IT WORKS

Isolate the training effect, convert it to money, and divide by the full cost.

LIMITS AND CRITIQUE

Isolating the effect is hard; ROI figures are often more precise than the evidence allows.

IN PRACTICE

Before claiming ROI for a learning platform, show that behavior changed; completion alone is not evidence.

EXAM LENS

Critique ROI through the counterfactual problem from Week 3: what would have happened without the training?

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

"Completion rates are the most misleading number in corporate learning." Critically evaluate using Kirkpatrick, its critics and transfer of training research. **20 marks**

A strong answer: explains the four levels; uses Alliger et al. and Holton; applies Baldwin and Ford and Blume et al.; proposes behavior-level measures.

Using learning science, design principles for a mobile learning program for frontline employees, and evaluate the Infosys Lex model's transferability. **20 marks**

A strong answer: applies spacing, retrieval, segmenting and SDT; distinguishes LMS, LXP and skills platforms; critiques transfer from graduates on laptops to frontline staff.

Why might points and badges reduce learners' motivation? **10 marks**

A strong answer: explains self-determination theory and the overjustification evidence.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **Learning systems should change behavior, not count completions**
   Measure transfer directly.

2. **Design for how memory works**
   Space, retrieve, interleave and segment.

3. **Evidence before ROI**
   No behavior change, no return to claim.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Case: Infosys Lex, digital reskilling at scale.
- NASSCOM FutureSkills (latest), selected sections.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Alliger, G. M., Tannenbaum, S. I., Bennett, W., Traver, H., & Shotland, A. (1997). A meta-analysis of the relations among training criteria. *Personnel Psychology, 50*(2), 341–358. https://doi.org/10.1111/j.1744-6570.1997.tb00911.x

Baldwin, T. T., & Ford, J. K. (1988). Transfer of training: A review and directions for future research. *Personnel Psychology, 41*(1), 63–105. https://doi.org/10.1111/j.1744-6570.1988.tb00632.x

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. P. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185–205). MIT Press.

Blume, B. D., Ford, J. K., Baldwin, T. T., & Huang, J. L. (2010). Transfer of training: A meta-analytic review. *Journal of Management, 36*(4), 1065–1105. https://doi.org/10.1177/0149206309352880

Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin, 132*(3), 354–380. https://doi.org/10.1037/0033-2909.132.3.354

Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Plenum.

Sources for this lecture (continued)

Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin, 125*(6), 627–668. https://doi.org/10.1037/0033-2909.125.6.627

Ebbinghaus, H. (1885). *Über das Gedächtnis: Untersuchungen zur experimentellen Psychologie*. Duncker & Humblot.

Holton, E. F., III (1996). The flawed four-level evaluation model. *Human Resource Development Quarterly, 7*(1), 5–21. https://doi.org/10.1002/hrdq.3920070103

Infosys. (2018, September 20). *Infosys launches Wingspan, a transformational learning solution for enterprises* [Press release].

Kirkpatrick, D. L. (1959). Techniques for evaluating training programs. *Journal of the American Society of Training Directors, 13*(11), 3–9.

Kirkpatrick, J. D., & Kirkpatrick, W. K. (2016). *Kirkpatrick's four levels of training evaluation*. ATD Press.

Knowles, M. S. (1984). *Andragogy in action: Applying modern principles of adult learning*. Jossey-Bass.

Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development*. Prentice Hall.

Sources for this lecture (continued)

Lombardo, M. M., & Eichinger, R. W. (1996). *The career architect development planner*. Lominger.

Mayer, R. E. (2009). *Multimedia learning (2nd ed.)*. Cambridge University Press.

Pashler, H., McDaniel, M., Rohrer, D., & Bjork, R. (2008). Learning styles: Concepts and evidence. *Psychological Science in the Public Interest, 9*(3), 105–119. https://doi.org/10.1111/j.1539-6053.2009.01038.x

Phillips, J. J. (1997). *Return on investment in training and performance improvement programs*. Gulf Publishing.

Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science, 17*(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

Rohrer, D., & Taylor, K. (2007). The shuffling of mathematics problems improves learning. *Instructional Science, 35*(6), 481–498. https://doi.org/10.1007/s11251-007-9015-8

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist, 55*(1), 68–78. https://doi.org/10.1037/0003-066x.55.1.68